

Debremarkos University

Computer Science

Web Programming

(CoSc3082)

Chapter: Two

Hyper Text Markup Language

(HTML)

Outline

Chapter 2: Hyper Text Markup Language (HTML) (10hr)

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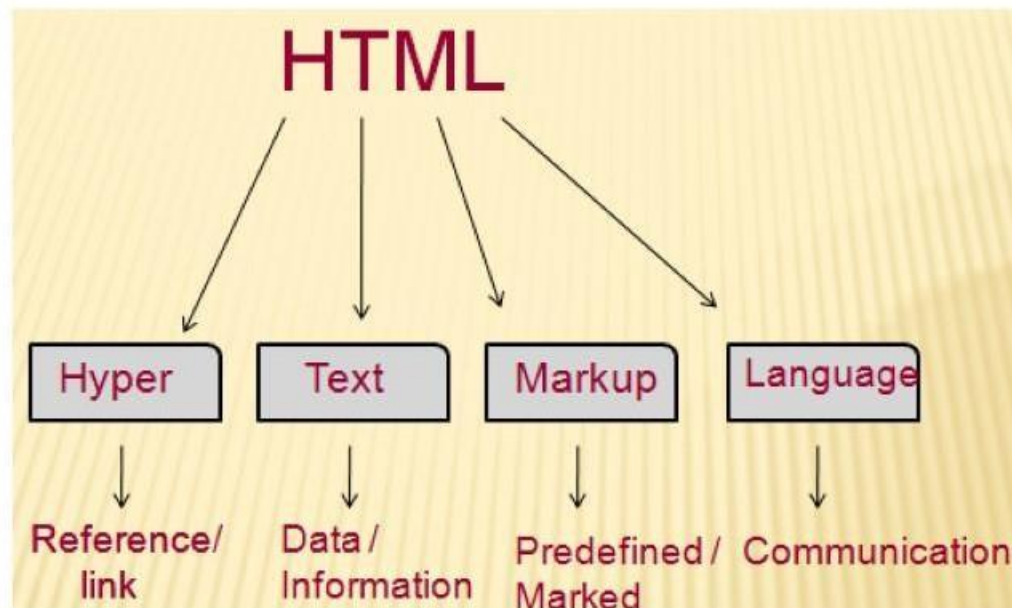
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What is a markup language?

- A markup language is a type of language used to **define and format the structure and presentation of text in documents**.
- It uses a **set of predefined tags or symbols to instruct the system on how to display the content**.
- Markup languages are commonly used in **web development, document creation, and data storage**.
- A markup language is a language that annotates text so that the computer can manipulate the text.
- The markup is the instructions for displaying or printing the text.
- Markup doesn't have to be computer readable. Annotations done in print or in a book are also markup.
- **Example:** Many students in school will highlight certain phrases in their text books.
- This indicates that the highlighted text is more important than the surrounding text. The highlight color is markup.

- Markup becomes a language when rules are codified around how to write and use the markup.
- That same student could have their own “note taking markup language” if they codified rules like “purple highlighter is for definitions, yellow highlighter is for exam details, and pencil notes in the margins are for additional resources.” But most markup languages are defined by an outside authority for use by many different people.



- '**Hypertext**' refers to the **linking of text with other documents**, while '
- **markup language**' denotes a language that utilizes a specific set of tags.

'Hypertext Markup Language'

- 'Hypertext' refers to the linking of text with other documents, while ‘
- **markup language**' denotes a language that utilizes a specific set of tags.
- Thus, HTML is the practice of displaying text, graphics, audio, video etc. in a certain way using special tags.
- **Note:** Tags are meaningful texts enclosed in angle braces, like '<...>'. For example, the '<head>' tag.
- Each tag has a unique meaning and significance in building an HTML page, and it can influence the web page in various ways.

HTML

- HTML (HyperText Markup Language) was created by **Tim Berners-Lee in 1991** as a **standard for creating web pages**.
- It's a **markup language** used to **structure content on the web**, defining elements like headings, paragraphs, links, and images.
- HTML is the **language of the web**, used to create websites.
- HTML consists of a **set of tags contained within an HTML document**, and the associated files typically have either a **".html" or ".htm"** extension.
- There are several versions of **HTML**, with **HTML5 being the most recent version**.
- HTML is like the skeleton of a website. It's a set of instructions that tells a web browser how to display text, images, videos, and other elements on a webpage.



Car Skeleton (only
body) is **HTML**

Car Painted or
Decorated is **CSS**

Car Engine and
Internal logic is **JS**

What You Can Do with HTML

Using HTML you can do the following .

- **publish documents** online with text, images, lists, tables, etc.
- **Can access web resources** such as images, videos or other HTML document via hyperlinks.
- **create forms to collect user inputs** like name, e-mail address, comments, etc.
- **include images, videos, sound clips, flash movies**, applications and other HTML documents directly inside an HTML document.
- create offline version of your website that work without internet.
- You can find the current location of your website's visitor.

Basics of HTML

Write HTML Using Notepad or any Text Editor.

Step 1: create html file (Notepad, Notepad ++, sublime Microsoft FrontPage, Macromedia Dreamweaver, Netscape Composer, Visual Studio

Step 2: Write Some HTML code

Step 3: Save the HTML code

Name the file "**index.htm/html**" and set the encoding to **UTF-8** (which is the preferred encoding for HTML files).

- HTML documents are files that end with a *.html* or *.htm* extension.

Step 4: View the HTML Page in Your Browser.

- The browser reads the HTML file and renders its content so that internet users can view it.

The structure of an HTML element



1. Identifies the document as written in HTML.
2. The head provides information about the document.
3. A descriptive title is required.
4. The body contains the content that displays in the browser

- Entire document enclosed within `<html>` and `</html>` tags

Two subparts:

Head

Enclosed within `<head>` and `</head>`

Within the head, more tags can be used to specify title of the page, meta-information, etc.

Body

Enclosed within `<body>` and `</body>`

Within the body, content is to be displayed

Other tags can be embedded in the body

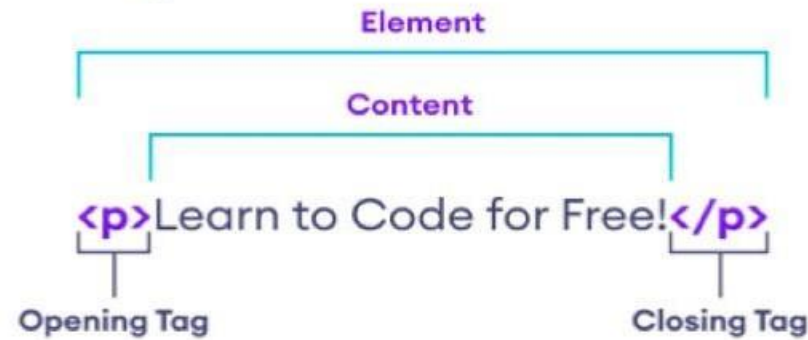
test.html

```
<!DOCTYPE HTML>
<html>
  <head>
    <title>My First HTML Page</title>
  </head>
  <body>
    <p>This is some text...</p>
  </body>
</html>
```



HTML Elements and tags

- HTML is comprised of “**elements**” and “**tags**”
- An HTML element is an **individual component of an HTML document**.
- Most HTML elements are written with a **start tag (or opening tag)** and an **end tag (or closing tag)**, with **content in between**.
- Syntax: **<tagname>content</tagname>**
- For example,



- The **opening tag**: This consists of the element name, **wrapped in angle brackets**.
 - It indicates the **start of the element** and the point at which the element's effects begin.
- The **closing tag**: This is the same as the opening tag, but with a **forward slash** before the element name.
 - It indicates the **end of the element** and the point at which the element's effects stop.
- **The content**: This is the content of the element, which **can be text, other elements, or a combination of both**.
- **The element**: The opening tag, the closing tag, and the content together make up the element.

HTML Tags

- Html tags are used to mark the start or end of an element,
- tag is composed of a keyword, surrounded by **angle brackets**, such as <html>, <head>, <body>, <title>, <p>, and so on.
- **tags normally** come in **pairs** like <html> and </html>.
- The first tag in a pair is often called the **opening tag** (or start tag), and the second tag is called the **closing tag** (or end tag).
- The markup tags tell the Web browser how to display the page
- Syntax **<tag> content </tag>**

When a web browser reads an HTML document, browser reads it from top to bottom and left to right. HTML tags are used to create HTML documents and render their properties.

- **Self-Closing Tag:** Used for empty elements that don't have any content (e.g., <tagname />).
- **
 Tag:** br stands for break line, it breaks the line of the code.
- **<hr> Tag:** hr stands for Horizontal Rule. This tag is used to put a line across the webpage.
- A **tag** is a part of the HTML syntax **used to create elements**.
- An **element** is the **combination of the opening tag, content, and closing tag** that forms a part of the **HTML document's structure**.

➤ What are HTML Attributes?

- HTML elements can have attributes, which provide **additional information about the element**.
- They are specified in the opening tag of the element and take the form of name-value pairs.
- **properties** of the element consists of **name/value** pairs like name="value".
- Attribute values should always be **enclosed in quotation marks**.

Attribute Name	Description
<u>id</u>	A unique identifier for an HTML element, used for styling or JavaScript interaction.
<u>class</u>	Defines one or more class names for an element, used for styling and applying CSS rules.
<u>src</u>	Specifies the source URL for external resources like images, audio, or video.
<u>href</u>	Specifies the URL of the linked resource, typically used in anchor (a) elements for hyperlinks.
<u>alt</u>	Provides alternative text for images, displayed if the image cannot be loaded or is unavailable.

Basic HTML Tags

0. html
1. Headings
2. Paragraph
3. Colors
4. Fonts
5. List
6. Anchor Tag
7. Image
8. Table
9. Form

0) HTML Tag

- The <html> tag tells the browser that this is an HTML document.
- The <html> tag represents the root of an HTML document.
- The <html> tag is the container for all other HTML elements

➤ `<html>`
 `<head>`
 `<title>`Title of the document`</title>`
 `</head>`

 `<body>`
 The content of the document.....
 `</body>`

 `</html>`

head tag

- The `<head>` in HTML is a section of the document that contains metadata and information about the page, rather than the content that is displayed directly on the page.
- It sits at the top of your HTML file, before the `<body>` tag.

```
<html>
```

```
<head>
```

```
<meta charset="UTF-8">
```

```
<title>My Web Page</title>
```

```
<meta name="description" content="A short description of my page">
```

```
<meta name="viewport" content="width=device-width, initial-scale=1.0">
```

```
<link rel="stylesheet" href="styles.css">
```

```
<script src="script.js" defer></script>
```

```
</head>
```

```
</body>
```

```
</html>
```

Common elements in <head>:

Tag	Purpose
<title>	Sets the title of the page (shown in browser tab)
<meta>	Provides metadata like character encoding, page description, keywords, etc.
<link>	Links external files like CSS stylesheets or icons
<style>	Allows internal CSS styling
<script>	Adds or links JavaScript functionality
<base>	Sets a base URL for all relative URLs in the page

Meta tag

- HTML lets you specify metadata - additional important information about a document in a variety of ways.
- they don't appear on the page itself but only in the page's source code.
- The META elements can be used to include name/value pairs describing properties of the HTML document, such as author, expiry date, a list of keywords, document author etc.
- The <meta> tag is used to provide such additional information.
- This tag is an empty element and so does not have a closing tag but it carries information within its attributes.
- place <meta> tags inside the header of the document which is represented by <head> and </head> tag
- A meta tag can have following attributes in addition to core attributes

No	Attribute & Description
1	Name: Name for the property. Can be anything. Examples include, keywords, description, author, revised, generator etc.
2	Content: Specifies the property's value.
3	Scheme: Specifies a scheme to interpret the property's value (as declared in the content attribute).
4	http-equiv Used for http response message headers. For example, http-equiv can be used to refresh the page or to set a cookie. Values include content-type, expires, refresh and set-cookie.

important usages of <meta> tag inside an HTML document

- **Meta Description Tag:** Provides a brief summary of the page content. It appears under the page title in search engine results.

`<meta name="description" content="This is an example of a meta description.">`

- **Meta Keywords Tag:** A list of keywords relevant to the page content.

`<meta name="keywords" content="HTML, CSS, JavaScript">`

- **Meta Charset Tag:** Specifies the character set for the HTML document.

`<meta charset="UTF-8">`

- **Meta Viewport Tag:** Controls the layout on mobile browsers, essential for responsive web design.

`<meta name="viewport" content="width=device-width, initial-scale=1.0">`

- **Meta Author Tag:** Provides information about the author of the document.

`<meta name="author" content="Abebe k">`

- **Meta Refresh Tag:** Redirects the page to another URL after a specified time.

`<meta http-equiv="refresh" content="30;url=https://example.com">`
redirects to <https://example.com> after 30 seconds.

- **Meta HTTP-Equiv Tag:** Provides HTTP headers to the browser.

`<meta http-equiv="Content-Type" content="text/html; charset=UTF-8">`

1) Headings

- A HTML heading or HTML **h tag** can be defined as a title or a subtitle which you want to display on the webpage.
- here are six different HTML headings which are defined with the `<h1>` to `<h6>` tags.
- `h1` is used for most important heading and `h6` is used for least important.

`<h1> text </h1>` -- largest of the six

`<h2> text </h2>`

`<h3> text </h3>`

`<h4> text </h4>`

`<h5> text </h5>`

`<h6> text </h6>` -- smallest of the six

Heading elements (h1....h6) should be used for headings only. They should not be used just to make text bold or big.

`align="position"` --left (default), center or right

HTML - Comments

- **Comment** is a piece of code which is **ignored by any web browser**.
- It is a **good practice to add comments into your HTML code**, **especially in complex documents**, to indicate sections of a document, and any other notes to anyone looking at the code. Comments help you and others understand your code and increases code readability.
- HTML comments are placed in between **<!-- ... -->** tags.
- So, any content placed with-in **<!-- ... -->** tags will be treated as comment and will be completely ignored by the browser.

```
<!-- This is a comment -->
```

```
<p>This is a paragraph.</p>
```

```
<!-- Remember to add more information here -->
```

Paragraph Tag

- To create well-structured text in your HTML document, the `<p>` tag is essential for defining paragraphs.
- It's used to define blocks of text that are treated as separate paragraphs in a web page.
 - `<p>This is a paragraph of text.</p>`
- Browsers automatically add space (margin) before and after a `<p>` element.

If you put a lot of spaces inside the HTML p tag, browser removes extra spaces and extra line while displaying the page.

Adding line breaks

- the `<br>` tag is used to insert a line break within a text element, such as a `<p>` tag
- `<p>This is the first line.<br>This is the second line.</p>`
- `<br>` is a self-closing tag (no `</br>` needed).

HTML `<pre>` Tag: Preformatted Text

- The `<pre>` tag in HTML defines preformatted text - it preserves both spaces and line breaks exactly as written in the HTML code.
- Text inside a `<pre>` element is typically displayed in a fixed-width font

Paragraph Tag

HTML <pre> Tag: Preformatted Text

- The <pre> tag in HTML defines preformatted text - it preserves both spaces and line breaks exactly as written in the HTML code.
- Text inside a <pre> element is typically displayed in a fixed-width font

<pre>

This text will
appear exactly
as written,
with all spaces
and line breaks preserved.

</pre>

- <pre> preserves ALL formatting (spaces, tabs, line breaks)
-
 only forces line breaks within other elements
- <p> automatically collapses whitespace

HTML paragraphs always **start on a new line**. To demonstrate this, let's create a webpage with two paragraphs.

This problem is solved by `<br>`tag

Html Style Attributes

- Setting the style of an HTML element, can be done with the **style attribute**.
- The HTML style attribute has the following **syntax**:

style="property:value;"

- The *property* is a CSS property. The *value* is a CSS value.
- Remember:
 - ✓ Use the **style** attribute for styling HTML elements
 - ✓ Use **background-color** for background color
 - ✓ Use **color** for text colors
 - ✓ Use **font-family** for text fonts
 - ✓ Use **font-size** for text sizes
 - ✓ Use **text-align** for text alignment

3) Colors

HTML Background Color

➤ The **background-color** property defines the background color for an HTML element:

- many are predefined (red, blue, green, ...)
- all colors can be specified as a six character hexadecimal value: #RRGGBB
- #FF0000 – red
- #888888 – gray
- #00FF00 – green
- #000000 – black
 - Example:-
 - `<body style="background-color:lightgrey;">`
`<h1>This is a heading</h1>`
`<p>This is a paragraph.</p>`
`</body>`

Text Color

➤ The **color** property defines the text color for an HTML element:

Example

```
<h1 style="color:blue;">This is a heading</h1>
<p style="color:red;">This is a paragraph.</p>
```

4) Fonts

- The **font-family** property defines the font to be used for an HTML element.
- Example `<h1 style="font-family:verdana;">This is a heading</h1>`
`<p style="font-family:courier;">This is a paragraph.</p>`

Text Size

- The **font-size** property defines the text size for an HTML element:
Example

```
<h1 style="font-size:300%;">This is a heading</h1>  
<p style="font-size:160%;">This is a paragraph.</p>
```

Text Alignment

- The **text-align** property defines the horizontal text alignment for an HTML element:

Example

```
<h1 style="text-align:center;">Centered Heading</h1>  
<p>This is a paragraph.</p>
```

HTML text Formatting tags

- HTML provides several tags that you can use to **make some text on your web pages to appear differently** than normal text, for example, you can use the tag **** to make the **text bold**, tag **<i>** to make the text **italic**, tag **<mark>** to highlight the text, tag **<code>** to display a fragment of computer code, tags **<ins>** and **** for marking editorial insertions and deletions, and more.
- In HTML **styling**, we use the HTML **style attribute**.
- the most commonly used formatting tags

Tag	Description		
	Defines bold text	<abbr>	Defines an abbreviation
<big>	Defines big text	<acronym>	Defines an acronym
	Defines emphasized text	<address>	Defines an address element
<i>	Defines italic text	<bdo>	Defines the text direction
<small>	Defines small text	<q>	Defines a short quotation
	Defines strong text	<cite>	Defines a citation
<sub>	Defines subscripted text	<dfn>	Defines a definition term
<sup>	Defines superscripted text		
<ins>	Defines inserted text		
	Defines deleted text		

HTML Bold Formatting

- The HTML **** element defines **bold** text.
- Example: `<p>This text is normal.</p>`
`<p><b>This text is bold</b>.</p>`

HTML Italic and Emphasized Formatting

- The HTML **<i>** element defines *italic* text, without any extra importance
- example :- `<p>This text is normal.</p>`
`<p><i>This text is italic</i>.</p>`

HTML Marked Formatting

- The HTML **<mark>** element defines **marked** or **highlighted** text:
- Example:- `<h2>HTML <mark>Marked</mark> Formatting</h2>`

HTML Deleted Formatting

The HTML **** element defines **deleted** (removed) text.

Example:- `<p>My favorite color is <del>blue</del> red.</p>`

HTML Subscript Formatting

The HTML **<sub>** element defines **subscripted** text.

Example:-

`<p>This is <sub>subscripted</sub> text.</p>`

HTML Superscript Formatting

The HTML **<sup>** element defines **superscripted** text.

Example:- `<p>This is <sup>superscripted</sup> text.</p>`

- The **Marquee tag** is a non-standard HTML element which is used to scroll a image or text horizontally or vertically.
- **<marquee>**This is html marquee **</marquee>**

Underlined Text

- If you write anything within `<u>.....</u>` element, is shown in underlined text.
- example:`<p> <u>Write Your First Paragraph in underlined text.</u></p>`

HTML quotes are used to put a short **quotation on your website**. To do so, you need to use HTML `q` tag and HTML `blockquote` tag.

- `q` tag is used to **put small quotation**.
- write your text within `<q>.....</q>` tag.
 - `<p> Great quote on love and life.</p>`
 - `<p> Dr. Seuss once said : <q>Reality is finally better than your dreams.</q></p>`
- `blockquote` tag is used to **define a large quoted section**.
- put the entire text within `<blockquote>.....</blockquote>` tag.
- **`<abbr>` element** defines an **abbreviation or an acronym**.

Example

`<p>The <abbr title="World Health Organization">WHO</abbr> was founded in 1948.</p>`

`<cite> ..... </cite>`: Tells the browser that this is a citation.

Usually displayed in italics

6) HTML LINK (Hyperlinks)

- A link is a connection from one web resource to another.
- Links allow users to **move from one page to another**, on any server anywhere in the world.
- Links used to create hyperlinks, which are **links to other webpages, files, locations within the same page, email addresses, or any other URL**.
- A **hyperlink** is a text or an image you can click on, and jump to another document.
- The **<a> tag or anchor tag** defines a hyperlink, which is used to link from one page to another.
- The **href attribute** is used to define the **address of the file to be linked**. In other words, it **points out the destination page**.
- Syntax: `<a href="url">link text</a>`
- **URL - the destination of the link**
- The **link text** is the visible part
- **Example:** -`<a href="http://www.w3schools.com/html/">Visit our tutorial</a>`
- By default, links will appear as follow in most of the browsers:
 - An **unvisited link** is underlined and blue.
 - A **visited link** is underlined and purple.
 - An **active link** is underlined and red.

HTML Links - The target Attribute

- The **target** attribute specifies **where to open the linked document**.
- This example will open the linked document in a new browser window or in a new tab:

Example

```
<a href="http://www.w3schools.com/" target="_blank">Visit W3Schools!</a>
```

Target Value	Description
<i>_blank</i>	Opens the linked document in a new window or tab
<i>_self</i>	Opens the linked document in the same frame as it was clicked (this is default)
<i>_parent</i>	Opens the linked document in the parent frame
<i>_top</i>	Opens the linked document in the full body of the window
<i>framename</i>	Opens the linked document in a named frame

➤ Hypertext references can be Internal, Local, or Global.

- **Global/ External** :- Links to other domains outside of your site

`<a href="http://www.microsoft.com">`

- **Local/ Internal** :- link to the same web site is specified with a relative URL (without http://www....). Or Links to other pages within your domain.

See these `<a href="./index.html">` references `</a>`

- **Internal / Anchor** : Links to anchors on the current page or allow users to jump to a specific section of a current web page.

- ✓ Creating bookmarks is a two-step process: first add the id attribute on the element where you want to jump, then use that id attribute value preceded by the hash sign (#) as the value of the href attribute of the `<a>` tag,

- ✓ `<a href="#reference">` Reference Section. `</a>`

- ✓

```
<p><a href="#C4">Jump to Chapter 4</a></p>
<p><a href="#C">Jump to Chapter 10</a></p>
<h2 id="C4">Chapter 4</h2>
<p>This chapter explains Java script</p>
```

download Attribute

- When linking to a web resource, we can specify that the resource is to be downloaded by using the download attribute.
`<a href="document.pdf" download>Download PDF</a>`
- This creates a link that, when clicked, will prompt the user to download the file `document.pdf`.
- Example: `<a href="/files/logo.png" download>Download Image </a>`
- The **email link** opens the user's default mail client to make it easier for the user to send mail to a specific address. For example,
- `<a href="mailto:name@domain.com" > Send Mail </a>`
- we can create **call links** to trigger the user's call app with a phone number.
- For example, `<a href="tel:+521953774446"> Call Us </a>`

HTML image inserting tag

- Images can improve the design and the appearance of a web page.
- ** tag** embeds an image within the HTML web page.
- Syntax: ``
- Example : ``

➤ Attributes of image tag

- **src:** Specifies the **path to the image**. This can be a **relative path**, **absolute path**, or **URL**.
- ✓ the **src** attribute tells the browser to load the image from the **same directory** where the webpage is located.
- ✓ For example, ``
- ✓ if the image is stored in a **subdirectory**, the path in the src attribute would be written as ``
- ✓ If the image is **located on another server entirely** or if we want to use an image from the web, we can provide an absolute URL for the image.
- ✓ For example,
- ✓ ``

HTML image inserting tag

➤ Attributes of image tag

- ✓ **alt:** Provides **alternative text for the image** if it cannot be displayed..
- ✓ **width:** Sets the width of the image (optional).
- ✓ **height:** Sets the height of the image (optional).
- The **** tag is empty, it contains attributes only, and does not have a closing tag.
- Example:

image map

- An image map technique that allows different parts of a single image to function as separate clickable links to various destinations.
- Instead of the entire image linking to one URL, specific regions (called "hotspots") within the image can link to different web pages or resources.
- Each area can link to different URLs, making it possible to create interactive images.
- **Tags for Image Maps**

Tag	Description
	Displays the image and links it to the image map using the usemap attribute.
<map>	Defines the image map (a collection of clickable areas). contains one or more <area> elements, each defining a clickable region.
<area>	Defines each clickable area within the map. Each <area> specifies the shape (rectangular, circular, or polygonal) and the link it points to.

- **usemap is an attribute** in the tag that links the image to a <map> element, which defines the interactive areas on that image.

image map

- Syntax:
- ``
- `usemap="#mapname"` — connects the image to a `<map>` element with `name="mapname"`.
- The `#` is required because you're referencing an internal name (like an anchor link).

➤ Example

- ✓ ``
- ✓ `<map name="dmulogo">`
- ✓ `<area shape="rect" coords="34,44,270,350" alt="link1" href="link.html">`
- ✓ `<area shape="circle" coords="400,200,50" alt="Link2" href="https://www.w3schools.com">`
- ✓ `<area shape="poly" coords="500,100,550,150,500,200,450,150" alt="Link3" href="cslogo.jpg"></map>`

Create an Image Map

- ``
- **Image with Map:** The `usemap="#dmulogo"` attribute links this image to the map named dmulogo.
- **Image Map Definition:** `<map name="dmulogo">`: Starts the image map definition with the name dmulogo.
- `<area shape="rect" coords="34,44,270,350" alt="Link 1" href="link.html">`:
 - Defines a rectangular clickable area linking to link.html.
- **shape:** Specifies the **shape of the clickable area** (rect for rectangle, circle for circle, poly for polygon).
- **coords:** Specifies the coordinates for the area.
 - ✓ For **rect**: `x1,y1,x2,y2` (top-left and bottom-right corners).
 - ✓ For **circle**: `x,y,radius` (center and radius).
 - ✓ For **poly**: `x1,y1,x2,y2,x3,y3,...` (vertices of the polygon).
 - The first area is a rectangle linking to `link.html`, covering **coordinates from (34,44) to (270,350)**.
 - The second area is a circle linking to `https://www.w3schools.com`, **centered at (400,200) with a radius of 50**.
 - The third area is a polygon linking to `cslogo.jpg`, with vertices at (500,100), (550,150), (500,200), and (450,150).
- **alt:** Provides alternative text for the area.
- **href:** Specifies the link for the area.

Images (cont.)

```

```

<i>align=position</i>	Image/Text Placement
Left	Image on left edge; text flows to right of image
Right	Image on right edge; text flows to left
Top	Image is left; words align with top of image
Bottom	Image is left; words align with bottom of image
Middle	Words align with middle of image

5) List

➤ HTML lists are used to **present list of information in well formed and semantic way.**

1. Ordered List or Numbered List (ol)
2. Unordered List or Bulleted List (ul)
3. Description List or Definition List (dl)

Ordered List

- | | | | | |
|------------|------------|------------|--------------|--------------|
| 1. Block-A | a) Block-A | A. Block-A | i. Block-A | I. Block-A |
| 2. Block-B | b) Block-B | B. Block-B | ii. Block-B | II. Block-B |
| 3. Block-C | c) Block-C | C. Block-C | iii. Block-C | III. Block-C |
| 4. Block-D | d) Block-D | D. Block-D | iv. Block-D | IV. Block-D |

Unordered List

- | | | |
|-----------|-----------|-----------|
| – Block-A | ○ Block-A | ▪ Block-A |
| – Block-B | ○ Block-B | ▪ Block-B |
| – Block-C | ○ Block-C | ▪ Block-C |
| – Block-D | ○ Block-D | ▪ Block-D |

5.1 ordered list

- Used to create a list of related items, in a specific order.
- An ordered list starts with the `<ol>` tag. Each **list item** starts with the `<li>` tag.
- The list items will be marked with numbers: **Example:**

```
<ol>
  <li>web programming</li>
  <li>compiler design</li>
  <li>intro to AI</li>
  <li>Economics</li>
</ol>
```

1. c++
2. php
3. html

1. web programming
2. compiler design
3. intro to AI
4. Economics

i. links
ii. image
iii. list
iv. formating tag

Ordered HTML Lists - The Type Attribute

- A **type** attribute can be added to an **ordered list**, to define the type of the marker:

Type	Description
type="1"	The list items will be numbered with numbers (default)
type="A"	The list items will be numbered with uppercase letters
type="a"	The list items will be numbered with lowercase letters
type="I"	The list items will be numbered with uppercase roman numbers
type="i"	The list items will be numbered with lowercase roman numbers

subOrdered List

```
<ol>
  <li> Item one </li>
  <li> Item two </li>
  <ol type="I" >
    <li> Sublist item one </li>
    <li> Sublist item two </li>
    <ol type="i">
      <li> Sub-sub list item one </li>
      <li> Sub-sub list item two </li>
    </ol>
  </ol>
</ol>
```

Types:

Type = 1 (default)

Type = a

Type = A

Type = I

Type = i

Output

1. Item one
2. Item two
 - I. Sublist item one
 - II. Sublist item two
 - i. Sub-sub list item one
 - ii. Sub-sub list item two

Unordered HTML Lists

- Used to create a list of related items, in no particular order.
- An unordered list starts with the **** tag. Each list item starts with the **** tag.
- The list items will be marked with bullets (small black circles)

Example

```
<ul> <li>OOP
```

```
<ul> <li>abstraction</li>
```

```
    <li>encapsulation</li></ul> </li>
```

```
<li>python</li>
```

```
<li>c++</li>
```

```
</ul>
```

- OOP
 - abstraction
 - encapsulation
- python
- c++

Unordered HTML Lists - The Style Attribute

- A **style** attribute can be added to an **unordered list**, to define the style of the marker:

Style	Description
list-style-type:disc	The list items will be marked with bullets (default)
list-style-type:circle	The list items will be marked with circles
list-style-type:square	The list items will be marked with squares
list-style-type:none	The list items will not be marked

5.2) Unordered List

```
<ul>
  <li> One </li>
  <li> Two </li>
  <ul type="circle">
    <li> Three </li>
    <li> Four </li>
  <ul type="square">
    <li> Five </li>
    <li> Six </li>
  </ul>
</ul>
</ul>
```

Types:

Type = disc (default)

Type = circle

Type = square

Output

- One
- Two
 - Three
 - Four
 - Five
 - Six

HTML Description Lists

- HTML also supports description lists.
- Used to create a list of terms and their descriptions.
- The **<dl>** tag defines the **description list**, the **<dt>** tag **defines the term** (name), and the **<dd>** tag **describes each term**:
- Example

HTML

- is a markup language used to create web page

CSS

- CSS is a stylesheet language that describes the presentation of an HTML

<dl>

<dt>HTML</dt>

<dd>- is a markup language used to create web page</dd>

<dt>CSS</dt>

<dd>- CSS is a stylesheet language that describes the presentation of an HTML</dd>

</dl>

HTML Tables

- HTML table allows you to **arrange data into rows and columns**.
- They are commonly used to **display tabular data** like **product listings**, **customer's details**, **financial reports**, and so on.
- Tables are defined with the **<table>** tag.
- Tables are divided into **table rows** with the **<tr>** tag.
- Table rows are divided into **table data** with the **<td>** tag.
- A table row can also be divided into **table headings** with the **<th>** tag.
- **<caption>** It defines the table caption.

An HTML Table with a Border Attribute

- If you do not specify a border for the table, it will be displayed without borders.
- A border can be added using the border attribute:

HTML Tables

- A border can be added using the border attribute:
- Example:-

```
<table style="width:50%" border="1">  
<caption>course with cr hr</caption>  
<tr>  
  <th>course name</th>  
  <th>course code</th>  
  <th>cr hr</th>  
</tr><tr>  
  <td>web programming</td>  
  <td>CoSc31</td>  
  <td>4</td>  
</tr>  
</table>
```

course name	course code	cr hr
web programming	CoSc31	4
compiler design	CoSc3062	5
Intro to AI	CoSc3063	5
Economics	CoSc3091	5

Colspan and Rowspan Attributes

- use **colspan** attribute if you want to To make a cell span more than one column
- use **rowspan** if you want to To make a cell span more than one row,

Table Cells that Span Many Columns

```
➤ Example:-  
➤ <tr>  
    <th>Name</th>  
    <th colspan="2">Telephone</th>  
</tr>  
<tr>  
    <td>Abebe</td>  
    <td>09537744</td>  
    <td>0953774468</td>  
</tr>
```

Name	Telephone	
Abebe	09537744	0953774468

Table Cells that Span Many rows

```
<table style="width:50%" border="1">  
  <tr>  
    <th>Name:</th>  
    <td>Abebe</td>  
  </tr>  
  <tr>  
    <th rowspan="2">Telephone:</th>  
    <td>095 77 854</td>  
  </tr>  
  <tr>  
    <td>0910 77 855</td>  
  </tr>  
</table>
```

Name:	Abebe
Telephone:	095 77 854
	0910 77 855

HTML Frames

- ❖ HTML frames are used to divide your browser window into multiple sections where each section can load a separate HTML document.
- ❖ A collection of frames in the browser window is known as a **frameset**.
- ❖ Frames are a way to divide a web page into multiple sections, each of which can load a separate HTML document.
- ❖ This allows for multiple documents to be displayed on a single screen, which can be useful for creating complex web interfaces.
- ❖ The window is divided into frames in a similar way the tables are organized: into rows and columns.

wellcome to web programming class

implemenintg html forms

text input control

First name

middle name

text input control password

password

multiline text input control textarea

to create textarea use textarea tag

describe about ur self

write something about ur self

radio box

sex

☒ male

☐ female

☐ other

types of link

[Send Mail click to send mail Call Us](#)

1. global link

[html](#)

[css](#)

[js](#)

2. local link

[html sample test](#)

3. internal link

[Jump to Chapter 6](#)

[Jump to Chapter 15](#)

Chapter 1

This chapter explains about web technology

Chapter 2

This chapter explains basics of html

Chapter 3

This chapter explain CSS

Chapter 4

well come to for web prog class

HTML Ordered List

course taken in these semester

1. web programming
2. compiler design
3. intro to AI
4. Economics

HTML Nested Ordered List

1. c++
2. php
3. html
 - i. links
 - ii. image
 - iii. list
 - iv. formating tag

HTML Unordered List

- OOP
 - abstraction
 - encapsulation
- python
- c++

HTML Frames Basic Structure

1. Frameset Element: Defines the layout of the frames.

2. Frame Element: Specifies each frame within the frameset.

3. NoFrames Element: Provides alternative content for browsers that do not support frames.

```
<frameset cols="50%,50%">
```

```
<frame src="frame1.html">
```

```
<frame src="frame2.html">
```

```
</frameset>
```

- Attributes of `<frameset>` and `<frame>`:

- **src:** Specifies the URL of the document to be displayed in the frame.
- **name:** Assigns a name to the frame, which can be targeted by links.
- **cols and rows:** Define the number and size of columns and rows in the frameset.

Example

```
<frameset cols = "30%, 40%, 30%">  
  <frame name = "top" src = "form.html" />  
  <frame name = "main" src = "link.html" />  
  <frame name = "bottom" src = "list.html" />  
  <noframes>  
    <body>The browser you are working does  
    not support frames.</body>  
  </noframes>  
</frameset>
```

Attributes of Frameset tag:

- **Column:** The size or width of the column is set in the frameset in the following ways:
 - ✓ **Use absolute value in pixel**
 - ✓ Example:
 - ✓ `<frameset cols = "300, 400, 300">`
 - ✓ **Use percentage value**
 - ✓ Example `<frameset cols = "30%, 40%, 30%">`
 - ✓ **Use wild card values:**
 - ✓ Example: `<frameset cols = "30%, *, 30%">`
 - ✓ In the above example * will take the remaining percentage for creating vertical frame.
- **rows:** The rows attribute is used to create horizontal frames in web browser.
- This attribute is used to define no of rows and its size inside the frameset tag.
 - ✓ The size of rows or height of each row use the following ways:
 - ✓ **Use absolute value in pixel**
 - ✓ Example: `<frameset rows = "300, 400, 300">`
 - ✓ **Use percentage value**
 - ✓ Example: `<frameset rows = "30%, 40%, 30%">`
 - ✓ **Use wild card values**
 - ✓ Example: `<frameset rows = "30%, *, 30%">`
 - ✓ In the above example * will take the remaining percentage for creating horizontal frame.

HTML Forms

- HTML Forms are required to collect different kinds of user inputs, such as contact details like name, email address, phone numbers, or details like credit card information, etc.
- Forms contain special elements called **controls** that users can interact with like **inputbox**, **checkboxes**, **radio-buttons**, **submit buttons**, etc.
- Users generally complete a form by modifying its controls
- e.g. **entering text**, **selecting items**, etc. and **submitting this form** to a web server for further processing.

Enter name:	
Enter password:	
Enter Email:	
Enter Gender:	☐ male ☐ female
Select Language:	Html ▼
register	

HTML Forms

- The **<form>** element defines an HTML form:
- it is main container for the form
- **Inside form element**, you place form controls such as text fields, radio buttons, and submit buttons.

syntax

```
<form action="server url" method="get|post">  
  //input controls e.g. textfield, textarea, radiobutton, button  
</form>
```

- **Example:** `<form action="action_page.php" method="post">` or `<form action="action_page.php" method="get">`
- **action attribute:** Specifies where to send the form data when the form is submitted.
- **Method attribute :** Defines the HTTP method to use when sending form data. Common values are GET and POST
- **method attribute** specifies the HTTP method (**GET** or **POST**) to be used when submitting the forms:

Use GET (the default method):

- If the form submission is passive (like a search engine query), and **without sensitive information** and When you use GET, the form data will be **visible in the page address**:
- suitable for **non-sensitive data**.

Use POST

- If the form is updating data, or includes **sensitive information (password)**.
- POST offers **better security because** the submitted data is not visible in the page address.

- **<input> tag:** used to define input fields in a form
 - several types of input fields
`{textbox, listbox, checkbox, radio, button, ...}`
 - **Attributes of input tag:**
 - `name="name"`
 - `type="type"`
`{text, hidden, password, file, submit, reset, button, checkbox, radio, image}`
 - `value="value"`
 - `disabled="disabled"`
 - `checked="checked"`
- `<input type="text" name="lastname">`**

Text Input controls

- There are three types of text input used on forms
- They are created using HTML **<input>** tag.

A. Single-line text input controls – This control is used for items that require only one line of user input, such as **search boxes or names**.

- ✓ **<input type="text">** defines a one-line input field for text input

Example: `<input type="text" name="inputname"/>`

First name:

B. Password input controls – This is also a single-line text input but it masks the character as soon as a user enters it. Example:

`<input type="password" name="inputname"/>`

C. Multi-line text input controls – This is used when the user is required to give details that may be longer than a single sentence. Multi-line input controls are created using HTML **<textarea>** tag.

Example :

`<textarea rows="8" cols="30">`

Deabre markos university becomes the lead

`</textarea>`

Description:

Radio buttons `<input type="radio">` defines a radio button.

- Radio buttons let a user **select ONE of a limited number of choices**:

– Example:

`<input type="radio" name="gender">` Male

☒ Male ☐ Female

`<input type="radio" name="gender">` Female

Check boxes let a user **select more than ONE of a number of choices**

– Example:

`<input type="checkbox" name="Roll1">` Roll No 1 `<br/>`

`<input type="checkbox" name="Roll2">` Roll No 2 `<br/>`

`<input type="checkbox" name="Roll3">` Roll No 3 `<br/>`

☐ Roll No 1
☐ Roll No 2
☐ Roll No 3

Dropdown list

- **<select>** tag is used to create a drop-down list in HTML.
- **<option>** tags inside the <select> tag define the available options in the list.
- Example:

```
<select name="programming">  
  <option value="c">C</option>  
  <option value="C++">C++</option>  
  <option value="OOP">OOp with Java</option>  
  <option value="java">Advanced programming</option>  
  <option value="python">python</option>  
</select>
```

The email input type allows the user to enter e-mail address.

- To use the email input type, you simply set the type attribute of the `<input>` element to email.
- `<input type="email" id="email" name="email" required>`
- You can style the email input field for different validation states, when an value is entered using the `:valid`, `:invalid` or `:required`
- **File Select box**
 - The file fields allow a user to browse for a local file and send it as an attachment with the form data.
 - `<input type="file" name="upload" id="file-select">`
- **Range Input:** `<input type="range">`: Slider control for selecting a value from a range.
- **Color Input** `<input type="color">`: Color picker.
- **Date and Time Input**
 - `<input type="date">`: Date picker.
 - `<input type="time">`: Time picker.
 - `<input type="datetime-local">`: Date and time picker.

- **Submit Button**:- Submit button is used to submit the data to the form action url. Or send data
- **Reset**: restore all form elements to their initial state.
- **Button**: take some action as specified by JavaScript.

Example:-

```
<input type = "submit" name = "submit" value = "Submit" />  
    <input type = "reset" name = "reset" value = "Reset" />  
    <input type = "button" name = "ok" value = "OK" />
```



Grouping Form Data with <fieldset>

The <fieldset> element groups related data in a form.

The <legend> element defines a caption for the <fieldset> element.

syntax:-

```
<form action="action_page.php">  
  <fieldset>  
    <legend>Personal information:</legend>  
    other form element  
  </fieldset>  
</form>
```

Form Attributes

- Common attributes for form controls:
 - **name:** Identifies the field when submitting
 - **value:** Default/pre-selected value
 - **placeholder:** Hint text
 - **required:** Mandatory field
 - **disabled:** Disables the control
 - **readonly:** Prevents editing
 - **autofocus:** Automatically focuses on page load
 - **pattern:** Regex validation
 - **min/max:** For numeric/date inputs
 - **step:** Increment value for numbers/dates

Validation on html form

The value Attribute

- The **value** attribute specifies the **initial value** for an input field:
- `<input type="text" name="firstname" value="John">`

The **readonly** Attribute

- The **readonly** attribute specifies that the input field is read only (**cannot be changed**):

Example

First name:
 `<input type="text" name="firstname" value="John" readonly>`

The disabled Attribute

- The **disabled** attribute specifies that the input field is disabled.
- A disabled element is **un-usable and un-clickable**.
- Disabled elements will not be submitted.

Example

`<form action="">`

First name:

`<input type="text" name="firstname" value="John" disabled>`

The maxlength Attribute

- The **maxlength** attribute specifies the maximum allowed length for the input field:

Example

```
<form action="">
```

First name:


```
<input type="text" name="firstname" maxlength="10">
```

The required Attribute

- The **required** attribute is a boolean attribute.
- When present, it specifies that an input field must be filled out before submitting the form.
- The required attribute works with the following input types: text, search, url, tel, email, password, date pickers, number, checkbox, radio, and file.

Username: <input type="text" name="username" **required**>

Inserting Multimedia in HTML

- **Multimedia** on the web is sound, music, videos, movies, and animations.
- It can be almost anything you can hear or see, like images, music, sound, videos, records, films, animations, and more.
- Web pages often contain multimedia elements of different types and formats.
- The most common way to discover the type of a file, is to look at the file extension.
- Multimedia files have formats and different extensions like: .wav, .mp3, .mp4, .mpg, .wmv, and .avi.

Embedding video

- To show a video in HTML, use the `<video>` element:
- The `controls` attribute adds video controls, like play, pause, and volume.
- To start a video automatically use the `autoplay` attribute:
- Only MP4, WebM, and Ogg video are supported by the HTML standard
 - `<video width="320" height="240" controls>`
 - `<source src="F:/entertainm/music/gojo.mp4" type="video/mp4">`
 - Your browser does not support the video tag.
 - `</video>`



Embedding the YouTube Videos

Step 1: **Upload video:** Go to YouTube upload video page and follow the instructions to upload your video.

Step 2: **Creating the HTML Code to embed the video**

- **click the share button**, a share panel will open displaying some more buttons.
- Now **click on the Embed button**, it will generate the HTML code to **directly embed the video into the web pages**.
- Just copy and paste that code into your HTML document where you want to display the video and you're all set.
By default video embedded inside an iframe.

Multimedia Formats

Format	File	Description
MPEG	.mpg .mpeg	MPEG. Developed by the Moving Pictures Expert Group. The first popular video format on the web. Not supported anymore in HTML.
AVI	.avi	AVI (Audio Video Interleave). Developed by Microsoft. Commonly used in video cameras and TV hardware. Plays well on Windows computers, but not in web browsers.
WMV	.wmv	WMV (Windows Media Video). Developed by Microsoft. Commonly used in video cameras and TV hardware. Plays well on Windows computers, but not in web browsers.
QuickTime	.mov	QuickTime. Developed by Apple. Commonly used in video cameras and TV hardware. Plays well on Apple computers, but not in web browsers.
RealVideo	.rm .ram	RealVideo. Developed by Real Media to allow video streaming with low bandwidths. Does not play in web browsers.
Flash	.swf .flv	Flash. Developed by Macromedia. Often requires an extra component (plug-in) to play in web browsers.
Ogg	.ogg	Theora Ogg. Developed by the Xiph.Org Foundation. Supported by HTML.
WebM	.webm	WebM. Developed by Mozilla, Opera, Adobe, and Google. Supported by HTML.
MPEG-4 or MP4	.mp4	MP4. Developed by the Moving Pictures Expert Group. Commonly used in video cameras and TV hardware. Supported by all browsers and recommended by YouTube.

Embedding Audio

- The newly introduced HTML5 `<audio>` element provides a standard way to **embed audio in web pages**.
- source defined by the `src` attribute.
`<audio src="foo.wav" controls autoplay>`
Your browser does not support the `<audio>` element.
`</audio>`
- most commonly used audio formats are **ogg**, **mp3** and **wav**.
`<audio controls autoplay>`
`<source src="F:/entertainm/collections/buna.mp3" type="audio/mp3" />`
Your browser does not support the audio element.
`</audio>`
- The `<embed>` element is used to embed multimedia content into an HTML document.
- The following code fragment **embeds audio files** into a web page.
`<embed src="F:/entertainm/collections/music1.mp3">`
- The following code fragment **embeds a Flash video** into a web page.
`<embed src="/examples/video/blur.swf" width="400px" height="200px">`

❖ **Note:** Only MP4, WebM, and Ogg video are supported by the HTML standard.

❖ *Common Audio Formats*

Format	File	Description
MIDI	.mid .midi	MIDI (Musical Instrument Digital Interface). Main format for all electronic music devices like synthesizers and PC sound cards. MIDI files do not contain sound, but digital notes that can be played by electronics. Plays well on all computers and music hardware, but not in web browsers.
RealAudio	.rm .ram	RealAudio. Developed by Real Media to allow streaming of audio with low bandwidths. Does not play in web browsers.
WMA	.wma	WMA (Windows Media Audio). Developed by Microsoft. Plays well on Windows computers, but not in web browsers.
AAC	.aac	AAC (Advanced Audio Coding). Developed by Apple as the default format for iTunes. Plays well on Apple computers, but not in web browsers.
WAV	.wav	WAV. Developed by IBM and Microsoft. Plays well on Windows, Macintosh, and Linux operating systems. Supported by HTML.
Ogg	.ogg	Ogg. Developed by the Xiph.Org Foundation. Supported by HTML.
MP3	.mp3	MP3 files are actually the sound part of MPEG files. MP3 is the most popular format for music players. Combines good compression (small files) with high quality. Supported by all browsers.
MP4	.mp4	MP4 is a video format, but can also be used for audio. Supported by all browsers.

HTML “canvas”

- The HTML “canvas” element is used to **draw graphics via JavaScript**.
- The “**canvas**” **element** is only a container for graphics.
- One must use JavaScript to actually draw the graphics.
- **Canvas** has several **methods** for drawing paths, boxes, circles, text, and adding images.
- The canvas would be a **rectangular area** on an HTML page.
- By default the canvas element has **300px of width and 150px of height** **without any border and content**.

Browser Support

The numbers in the table specify the first browser version that fully supports the `<canvas>` element.

Element					
<code><canvas></code>	4.0	9.0	2.0	3.1	9.0

Core Functionalities of canvas

- The Canvas API offers a vast array of methods and properties for drawing and manipulating various visual elements.
- Here are some key functionalities:
- **Drawing shapes:** Create lines, rectangles, circles, ellipses, polygons, and more with customizable styles and colors.
- **Rendering text:** Display text with different fonts, sizes, styles, and colors, including effects like shadows and gradients.
- **Image manipulation:** Draw images directly onto the canvas, scaling, rotating, clipping, and applying filters for dynamic effects.
- **Animation:** Implement smooth animations through requestAnimationFrame, allowing for interactive experiences and visual storytelling.
- **User interaction:** Capture user input through event listeners, enabling interaction with the canvas content.

Understanding Canvas Coordinates

- The canvas is a two-dimensional rectangular area.
- The coordinates of the top-left corner of the canvas are (0, 0) which is known as origin, and the coordinates of the bottom-right corner are (*canvas width*, *canvas height*).
- Here's a simple demonstration of canvas default coordinate system.



Place your mouse pointer within the canvas area demonstrated above and you will get its current coordinates relative to the canvas.

Syntax:

```
<canvas>
```

```
  Content...
```

```
</canvas>
```

Canvas: Drawing graphics

- **The overview is split into two parts:**

1. Declaring an HTML5 canvas element.
2. Drawing graphics on the canvas element.

- **Declaring a Canvas Element**

- First, let's see how to declare a canvas element in an HTML page:

```
<canvas id="ex1" width="500" height="150"  
      style="border: 1px solid #cccccc;">
```

HTML5 Canvas not supported

```
</canvas>
```

- Drawing graphics on an HTML5 canvas element is done using **JavaScript**, following these steps:

1. Wait for the page to be fully loaded.
2. Obtain a reference to the canvas element.
3. Obtain a 2D context from the canvas element.
4. Draw graphics using the draw functions of 2D context

Canvas: Drawing graphics

Drawing graphics on an HTML5 canvas element is done using JavaScript, following these steps:

1.Wait for the page to be fully loaded.

```
window.onload = function() {  
    drawExamples();  
}
```

2.Obtain a reference to the canvas element.

```
var canvas = document.getElementById("ex1");
```

3. Obtain a 2D context from the canvas element.

```
var context = canvas.getContext("2d");
```

4. Draw graphics using the draw functions of 2D context.

```
context.fillStyle = "#ff0000";  
context.fillRect(10,10, 100,100);
```

create Canvas

- A single HTML page can **hold multiple canvas elements** but specifying the **ID** and the **height & width attribute** is a must to define the size of the canvas
- `<canvas id="myCanvas" width="200" height="100"></canvas>`
- Here is an example of a basic, empty canvas:



- Basic Canvas Example

```
<canvas id="myCanvas" width="200" height="100" style="border:1px solid #000000;">
</canvas>
```

Draw rectangle

- To draw a rectangle, we can use the **fillRect(x, y, width, height)** method to set the coordinates and the dimensions, along with the **fillStyle** property to set the fill color.

Drawing with JavaScript

```
<script>
var c = document.getElementById("myCanvas");
var ctx = c.getContext("2d");
ctx.fillStyle = "#FF0000";
ctx.fillRect(0,0,150,75);
</script>
```



Drawing Line on Canvas

- If you want to draw a straight line on the canvas, you can use the following two methods.
- **moveTo(x,y):** It is used to define the starting point of the line.
- **lineTo(x,y):** It is used to define the ending point of the line.
- If you draw a line which starting point is (0,0) and the end point is (200,100), use the stroke method to draw the line.
- **stroke()** Draws the line. The default stroke color is black
- `<canvas id="myCanvasLine" width="200" height="100" style="border:1px solid #d3d3d3;">`
- Your browser does not support the HTML5 canvas tag. `</canvas>`

`<script>`

```
var c = document.getElementById("myCanvasLine");
```

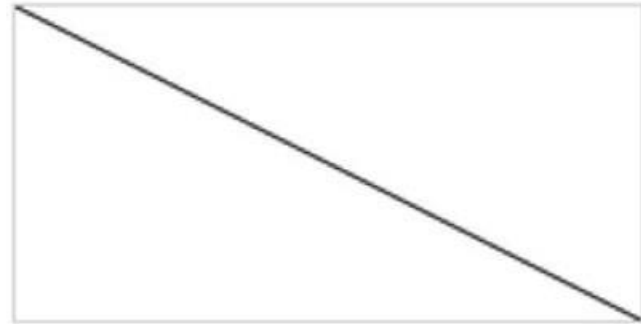
```
var cctx = c.getContext("2d");
```

```
cctx.moveTo(0,0);
```

```
cctx.lineTo(200,100);
```

```
cctx.stroke();
```

`</script>`



Drawing Circle on Canvas

- If you want to draw a circle on the canvas, you can use the **arc()** method:
- `arc(x, y, r, start, stop)`
- To sketch circle on HTML canvas, use one of the `ink()` methods, like `stroke()` or `fill()`.
- `<canvas id="myCanvasCircle" width="200" height="100" style="border: 1px solid #d3d3d3;">`

Your browser does not support the HTML5 canvas tag. `</canvas>`

`<script>`

`var c = document.getElementById("myCanvasCircle");`

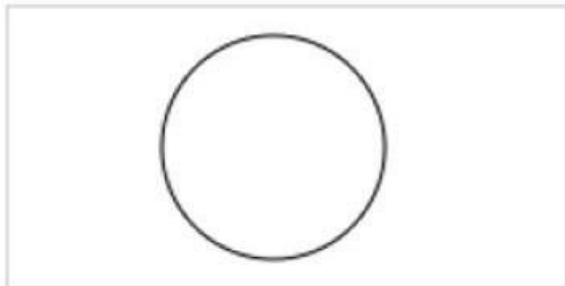
`var cctx = c.getContext("2d");`

`ctx.beginPath();`

`ctx.arc(95,50,40,0,2*Math.PI);`

`ctx.stroke();`

`</script>`



`arc()` - draws a circle where,
the circle is **centered at (95, 50)**
the circle has a **radius of 40 pixels**
the circle is drawn to cover the entire circumference in a counter-clockwise direction, by **starting at 0 radians and ending at 2*Math.PI radians**

Drawing text on canvas

- There are property and methods used for drawing text on the canvas.
- **font property:** It is used to define the font property for the text.
- **fillText(text,x,y) method:** It is used to draw filled text on the canvas. It looks like bold font.
- **strokeText(text,x,y) method:** It is also used to draw text on the canvas, but the text is unfilled.

```
<canvas id="myCanvasText2" width="300" height="100" style="border:1px solid #d3d3d3;">
```

Sorry! Upgrade your browser. It does not support the HTML5 canvas tag.</canvas>

```
<script>
```

```
var c = document.getElementById("myCanvasText2");
```

```
var cctx = c.getContext("2d");
```

```
cctx.font = "30px Arial";
```

```
cctx.strokeText("Hello JavaTpoint",10,50);
```

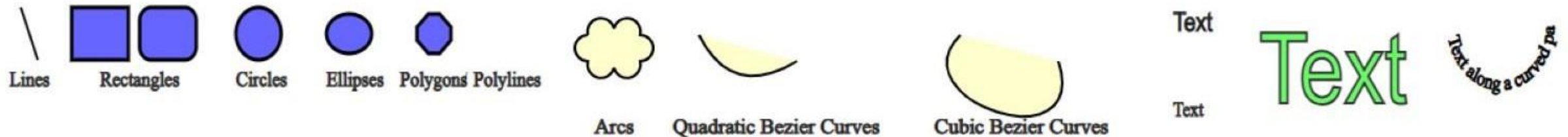
```
</script>
```



Hello JavaTpoint

SVG (Scalable Vector Graphics)

- SVG (Scalable Vector Graphics) is a **vector graphics format** used to create and display two-dimensional images on the web and other platforms.
- it is an XML-based markup language used for creating scalable 2D graphics and graphical applications.
- XML stands for **Extensible Markup Language** and it is a **data description language**.
- XML does **not have any predefined tags** hence; the users are required to define their own tags depending on the need.
- the image data is stored in a text-based format that can be easily edited, manipulated, and integrated with other web technologies like CSS and JavaScript
- SVG is a W3C recommendation.
- It is mostly useful for vector-type diagrams like **Pie charts, and two-dimensional graphs in an X, Y coordinate system.**
- The HTML <svg> element is **a container for SVG graphics.**
- SVG has several methods for drawing paths, boxes, circles, text, and graphic images.



How to use SVG in HTML?

- There are two ways of embedding the SVG files in HTML

- ✓ Using `<svg>` tag
- ✓ Using `iframe`
- ✓ Using `<img>` tag

- **Using `<svg>` tag**

HTML allows embedding SVG directly using `<svg>...</svg>` tag which has the following syntax `<svg>`

`<!-- code to create graphics -->`

`</svg>`

Example `<svg>`

`<circle cx="40" cy="40" r="24" style="stroke:#006600; fill:#00cc00"/>`

`</svg>`

- **Iframe element**

- Since the browsers can show SVG images if you type in the URL to the SVG image, you can also include an SVG image in your HTML page by using an `iframe`. Here is an example:

- `<iframe src="mySvgImage.svg" width="200" height="200" ><iframe src="phone-call.svg" width="800" height="500" >`

How to use SVG in HTML?

- **Using tag**

- ✓ We can directly embed the SVG files inside our web page using the `src` attribute of `<img>` tag as shown below.
- ✓ We can pass either the path or an online link to the svg image.
- ✓ `<img src = "laptop-computer-svgrepo-com.svg" height = "100px" width = "200px" />`

- **There are a few ways to include SVG content in an HTML document:**

- **Inline SVG:** You can `directly embed the SVG markup` within the HTML document, like this:

```
<svg width="100" height="100">  
  <circle cx="50" cy="50" r="40" fill="blue" />  
</svg>
```

- This allows for direct manipulation of the SVG elements using JavaScript and CSS
- **External SVG:** You can also link to an external SVG file using the `<img>` or `<object>` tags:
``
`<object type="image/svg+xml" data="my-graphic.svg"></object>`

Common Attributes of SVG Elements

Dimensions:

width and height: Define the dimensions of the SVG viewport.

viewBox: Defines the coordinate system and aspect ratio of the SVG content.

Shapes:

x, y, rx, ry: Attributes for positioning and rounding the corners of basic shapes like `<rect>`, `<circle>`, and `<ellipse>`.

Appearance:

fill, stroke, stroke-width: Attributes that define the color, border, and line thickness of shapes.

Transformations:

transform: Applies a transformation (e.g., translate, scale, rotate, skew) to an SVG element.

Events: onclick, onmouseover, onmouseout:

Text: `<text>`, font-family, font-size: Attributes for displaying and formatting text within an SVG.

Some Common Attributes of SVG Elements

The following table contains a list of attributes of SVG Elements –

S.No.	Attribute & Description
1	X The top-left x-axis coordinate.
2	Y The top-left y-axis coordinate.
3	width The width of figure.
4	height The height of figure.
5	rx The x-axis' roundness.
6	ry The y-axis' roundness.
7	style Indicate an inline style.

Embedding SVG in HTML:

- There are a few ways to include SVG content in an HTML document:
- **Inline SVG:** You can [directly embed the SVG markup](#) within the HTML document, like this:

```
<svg width="100" height="100">  
  <circle cx="50" cy="50" r="40" fill="blue" />  
</svg>
```

- This allows for direct manipulation of the SVG elements using JavaScript and CSS
- **External SVG:** You can also link to an external SVG file using the [](#) or [<object>](#) tags:

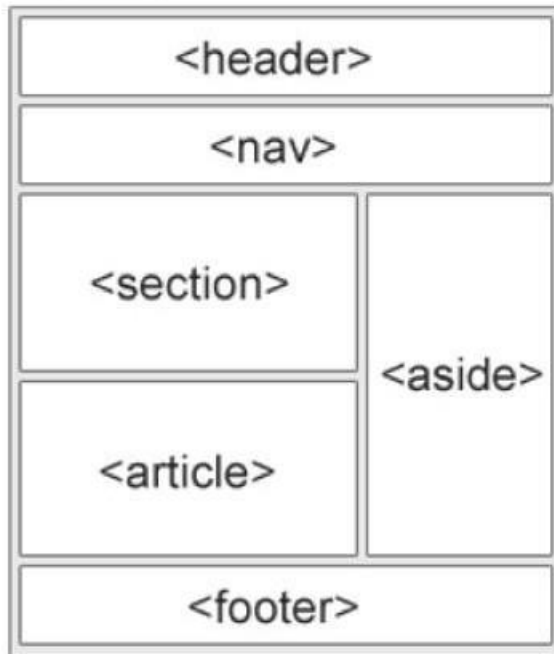
```
  
<object type="image/svg+xml" data="my-graphic.svg"></object>
```

Embedding SVG in HTML:

- **Common SVG Elements:**
- `<svg>`: The root element of an SVG document.
- `<circle>`: Draws a circle.
- Attributes: `cx`, `cy` (center coordinates), `r` (radius).
- `<rect>`: Draws a rectangle.
- Attributes: `x`, `y` (top-left corner coordinates), `width`, `height`.
- `<line>`: Draws a line.
- Attributes: `x1`, `y1` (start coordinates), `x2`, `y2` (end coordinates).
- `<polygon>`: Draws a shape with multiple sides.
- Attribute: `points` (coordinates of the vertices).
- `<path>`: Draws a complex shape described by a series of commands.
- Attribute: `d` (a string containing path data).

■ Creating Website Layouts

- Creating a website layout is the activity of positioning the various elements that make a web page in a well-structured manner and give appealing look to the website.
- Creating Website Layouts can be easily achieved by using the HTML tags, such as `<table>`, `<div>`, `<header>`, `<footer>`, `<section>`, etc. and adding some CSS styles to them.
- HTML5 offers new semantic elements that define different parts of a web page:



- `<header>` - Defines a header for a document or a section
- `<nav>` - Defines a container for navigation links
- `<section>` - Defines a section in a document
- `<article>` - Defines an independent self-contained article
- `<aside>` - Defines content aside from the content (like a sidebar)
- `<footer>` - Defines a footer for a document or a section
- `<details>` - Defines additional details
- `<summary>` - Defines a heading for the `<details>` element

■ Creating Website Layouts

web programming

HTML

CSS

JS

Creating website layouts

this involves the process of putting the contents such as text, images, and so on into rows and columns, layout is created using an HTML table with 3 rows and 2 columns –
the first and last row spans both columns using the table's colspan attribute:.

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■ Creating Website Layouts

■ HTML Table Based Layout

- , this involves the process of putting the contents such as text, images, and so on into rows and columns.
- The following layout is created using an HTML table with 3 rows and 2 columns — the first and last row spans both columns using the table's colspan attribute:

create website with table layout	
Home About Contact	Welcome to our site Here you will learn how to create websites...
copyright © G4 CS	

■ Creating Website Layouts

■ using Div Tag

- The HTML <div> tag is used to group the large section of HTML elements together.
- <div> tag is just like a container unit which is used to encapsulate other page elements and divides the HTML documents into sections.
- The div tag is generally used by web developers to group HTML elements together and apply CSS styles to many elements at once

```
<div class="container">
  <div class="header">
    <h1>web programming Tutorial 2016 </h1>
  </div>
  <div class="wrapper clearfix">
    <div class="nav">
      <ul>
        <li><a href="link.html">Html Link</a></li>
        <li><a href="ist.html">Html list</a></li>
        <li><a href="#">Contact</a></li>
      </ul>
    </div>
    <div class="section">
      <h2>Welcome to our site</h2>
      <p>Here you will learn how to create websites...</p>
    </div>
  </div>
</div>
```

Web Programming

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- [Html Link](#)
- [Html list](#)
- [Contact](#)

basic tutorials of html,css and javascript

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Thank You

Chapter 3